

38. THE BRAIN: THE VENTRICLES

DURATION : 14:52

STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the brain's ventricular system, integrating key embryological concepts such as telencephalisation, ascensus, and hemispheric differentiation. You will learn how the growth of the lateral ventricles and other brain structures influences not only neurological development but also the dynamics of osteopathic therapy. The formation of the choroid plexuses, as well as the circulation of cerebrospinal fluid, will also be addressed, offering valuable insights into the interactions between the brain and the rest of the body, such as connections with the peritoneum. This understanding will enrich your clinical approach and stimulate your practice.

THE BRAIN: THE VENTRICLES

INTRODUCTION TO THE VENTRICULAR SYSTEM

We will integrate the **ventricular system** into the developmental stages already studied. Segmentation and the different structures are essential. A lateral view of the image offers a better perspective.

It is crucial to integrate this developmental movement. When examining a patient, keep this global perspective in mind. Even when working on the feet, unsuspected connections, particularly with the peritoneum, can be discovered.

PROCESS OF ASCENSUS AND HEMISPHERISATION

We distinguish the **midbrain**, the **highbrain**, and the spinal cord. The process of ascensus is clearly visible, accompanied by hemispherisation.

Do not forget the overall growth and repositioning of the brain, which resembles a cabbage. Everything happens within this process.

ROLE OF NEURO-PORES AND PROSENCEPHALON

The anterior **neuro-pores** are planes of dynamisation. The complete closure of the inner and outer neuro-pores triggers a dynamic of brain development.

Initially, we essentially have the **prosencephalon**, with the anterior neuro-pore, followed by the mesencephalon and the rhombencephalon. This is a structure containing fluid inside this tube and these vesicles.

TELENCEPHALISATION AND VENTRICLE FORMATION

The process of **telencephalisation** gives rise to the **lateral ventricles**. This is how the third, fourth, and lateral ventricles are formed.

- **Initial vesicles:** Initially, three small vesicles containing fluid.
- **Brain growth:** The brain begins to grow, and we will explore the inside of the ventricles.

Imagine you are at the level of the prosencephalon, which is flexed, so the entire system leans forward.

CLOSURE OF NEURO-PORES AND LATERAL EXPANSION

The closure of the anterior neuro-pore and the posterior neuro-pore occurs between the 26th and 28th day, approximately one lunar cycle.

At that time, growth occurs with a **lateral expansion** of the prosencephalon. From a simple ball, it becomes two balls.

FLUID MOVEMENT AND HORN FORMATION

The **cerebrospinal fluid** follows this movement. It moves forward, then the system moves backward, thus forming the horns of the lateral ventricles.

This movement follows the development of the prosencephalon.

INTERVENTRICULAR FORAMEN AND TELENCEPHALISATION

The **interventricular foramen** is a small expansion that occurs simultaneously with the process of telencephalisation, on both sides. It follows the complete movement, a fulcrum, then returns backward.

This is solely the developmental movement of the telencephalon. Sometimes, one hemisphere develops faster than the other.

PRACTICE AND BRAIN EXPLORATION

During a practice, we will revisit this dynamic to explore the depth of the brain, as if swimming with whales, without touching them.

ORIGIN OF THE PROSENCEPHALON AND EYES

One could say that initially, the prosencephalon is the **third ventricle**. The eye is an expansion from it. (We will discuss this later.)

The eye is made of nervous tissue, not fluid. Its origin is an expansion of the third ventricle.

The interthalamic adhesion connects the two thalami, with fluid around them.

CEREBROSPINAL FLUID COMPARTMENTS

There is not only the intra-encephalic compartment but also the extra-encephalic compartment, whose metabolic role is increasingly being discovered.

CHOROID PLEXUS AND FLUID FLOW

The **choroid plexuses** are areas of fluid production, present in each ventricle: the lateral ventricles, the third, and the fourth ventricle. (The third ventricle is the first to form in this chronology.)

The lateral ventricles are an expansion that follows the development of the telencephalon.

- **Foramen of Monro:** Located between the third ventricle and the lateral ventricles.

POSTERIOR FLUID CIRCULATION

After the third ventricle, the **aqueduct of Sylvius** leads to the fourth ventricle.

- **Foramen of Magendie:** A significant expansion towards the brain, a main opening to the extracephalic compartment.

PACCHIONIAN GRANULATIONS

The **Pacchionian granulations** are resorption sites located along the sinus.

KEY DEVELOPMENTAL MOVEMENTS

Key movements include **cephalic flexion**, **cervical curvature**, **pontine curvature**, and especially this great movement of hemispherisation and telencephalisation that reshapes the embryo.

- **Initial structures:** Prosencephalon, mesencephalon, rhombencephalon.

- The green part is particularly important for understanding this developmental dynamic.

It will be interesting to review the days of development later to correlate this with digestive or other phenomena.

CONNECTIONS AND FULCRUMS

This recalls the different fulcrums and the general affection due to the great cephalisation.

It is also interesting to remember the connections with the tip of the **notochord**, the **supraocciput**, the **basiocciput**, and the **presphenoid**.

THE ASCENSUS CEREBRALI AND THE FORMATION OF THE FACE

The **ascensus cerebrale** is the process of cerebralisation and the formation of the face.

- **Face development:** Initially, we observe the nasal placode, the mouth, and the heart in the small embryo.
- **Fulcrum:** The growth of the telencephalon occurs with a crucial fulcrum at the level of the future **crista galli** or the **nasion**.

Everything returns towards the median axis of the embryo during growth.

TELENCEPHALISATION AND ANTERIOR MEDIAN LINE

The process of telencephalisation has a direct action on the formation of the **anterior median line**.

We observe enormous growth and all the attraction that operates.

CEREBRALISATION AND PALATE STRUCTURES

The process of **cerebralisation** (ascensus cerebrale) will influence all areas of the palate, which we will explain next time. There is a very important fulcrum here.

Telencephalic development leads to the approximation of the palatine, the flexion of the choanae, and the posterior resultant. This is a simultaneous change between brain development backward, corticalisation, and ventricularisation.

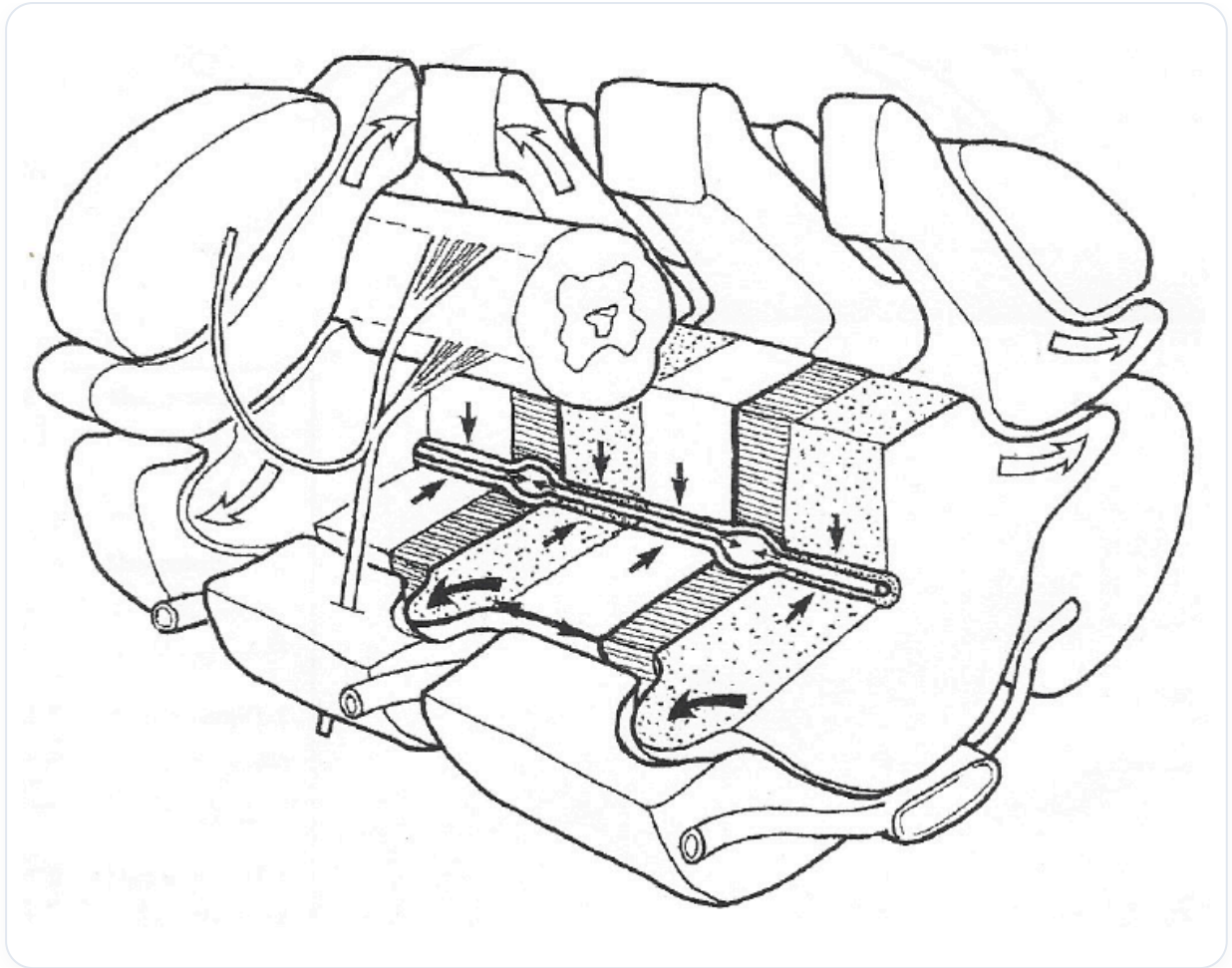


FIG. — *Spinal CSF circulation*